

HVERAVELLIR, Iceland

WMO: 041560

Lat: 64.8668N

Lon: 19.5622W

Elev: 641

StdP: 93.86

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-17.3	-15.2	-20.5	0.7	-16.8	-18.3	0.8	-14.9	26.9	-1.2	25.0	-2.3	4.3	340	1.062

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.3	16.1	10.4	14.1	9.5	12.5	8.8	11.4	14.6	10.3	13.0	9.5	11.8	5.4	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.7	8.1	12.4	8.9	7.6	11.3	8.0	7.2	10.4	34.2	14.6	31.6	13.1	29.6	11.8	15.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.2	19.4	17.0	DB	-20.5	19.0	1.6	2.7	-21.6	21.0	-22.5	22.6	-23.4	24.1	-24.6	26.1	(4)
(5)			WB	-20.8	13.4	1.6	1.5	-21.9	14.4	-22.9	15.3	-23.8	16.1	-24.9	17.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.1	-5.1	-5.8	-5.3	-2.1	1.8	6.2	8.4	7.5	4.0	-0.7	-3.4	-4.9	(6)	
	DBStd	6.36	4.40	4.65	4.46	3.92	3.24	2.92	2.58	2.38	3.38	3.88	4.26	4.76	(7)	
	HDD10.0	3644	469	442	474	364	253	117	64	84	182	331	403	461	(8)	
	HDD18.3	6663	727	675	733	614	511	364	309	337	431	589	653	719	(9)	
	CDD10.0	22	0	0	0	0	0	3	13	6	1	0	0	0	(10)	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	7.6	8.8	9.0	8.9	7.9	6.9	6.4	6.1	6.2	7.3	7.4	7.8	8.7	(14)	
(15) Precipitation	PrecAvg	1737	186	185	159	119	90	84	99	127	156	168	164	200	(15)	
	PrecMax	2159	346	337	281	254	169	177	179	260	251	398	321	313	(16)	
	PrecMin	1032	90	74	57	40	9	24	49	69	92	82	45	89	(17)	
	PrecStd	285	73	64	63	48	34	39	35	48	40	62	63	61	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.1	3.8	4.1	8.0	12.0	16.9	20.0	18.3	12.6	9.0	6.0	4.9	(19)	
		MCWB	3.0	2.8	2.9	5.8	7.1	9.9	12.5	11.7	9.7	7.5	4.8	3.7	(20)	
	2%	DB	2.7	2.6	2.8	5.0	9.8	14.5	16.8	15.1	10.7	7.2	4.6	3.6	(21)	
		MCWB	1.7	1.7	1.9	3.7	6.3	9.1	11.2	10.3	8.5	5.9	3.5	2.6	(22)	
	5%	DB	1.8	1.8	1.9	3.6	8.0	12.9	14.6	13.0	9.5	6.0	3.3	2.5	(23)	
		MCWB	0.9	1.1	1.2	2.4	5.3	8.3	9.9	9.3	7.8	4.9	2.4	1.5	(24)	
	10%	DB	0.9	0.8	0.9	2.7	6.4	11.3	12.8	11.5	8.5	4.9	2.3	1.5	(25)	
		MCWB	0.3	0.3	0.3	1.7	4.4	7.6	9.2	8.7	7.0	3.8	1.4	0.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.4	3.1	3.1	5.8	7.9	11.4	13.6	12.8	10.4	7.8	5.2	3.9	(27)	
		MCDB	4.0	3.7	3.7	7.7	11.0	15.4	19.0	15.6	12.0	8.9	6.0	4.4	(28)	
	2%	WB	1.9	1.9	2.1	3.9	6.7	9.8	11.7	11.0	9.0	6.1	3.7	2.8	(29)	
		MCDB	2.5	2.4	2.6	4.9	9.2	13.1	15.3	13.6	10.1	7.0	4.4	3.5	(30)	
	5%	WB	1.1	1.1	1.2	2.7	5.7	8.9	10.7	10.0	8.1	5.0	2.5	1.6	(31)	
		MCDB	1.7	1.7	1.8	3.5	7.6	12.1	13.7	12.3	9.3	6.0	3.2	2.3	(32)	
	10%	WB	0.2	0.3	0.3	1.8	4.6	7.9	9.7	9.1	7.1	3.9	1.5	0.7	(33)	
		MCDB	0.8	0.8	0.9	2.6	6.2	10.7	12.1	11.1	8.5	4.8	2.2	1.4	(34)	
(35) Mean Daily Temperature Range	MDBR		5.3	5.4	5.5	5.2	5.5	6.9	6.3	6.2	4.8	4.4	4.6	4.9	(35)	
	5% DB	MCDBR	4.8	5.0	4.9	4.8	7.9	9.8	9.8	9.2	5.7	5.3	4.7	4.7	(36)	
		MCWBR	4.4	4.6	4.4	3.9	5.0	5.5	5.1	5.1	4.1	4.6	4.2	4.1	(37)	
	5% WB	MCDBR	4.7	4.8	4.8	4.5	7.1	8.4	8.1	7.3	5.1	5.1	4.6	4.6	(38)	
		MCWBR	4.5	4.5	4.5	3.8	4.8	5.1	4.6	4.6	4.0	4.6	4.2	4.2	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.203	0.264	0.297	0.318	0.326	0.328	0.329	0.310	0.301	0.266	0.190	0.199	(40)	
	taud		1.863	2.083	2.125	2.159	2.245	2.339	2.387	2.440	2.397	2.198	1.892	1.840	(41)	
	Ebn at Noon		289	618	770	846	878	885	874	860	782	614	301	70	(42)	
	Edh at Noon		26	61	92	114	115	108	100	86	71	52	23	9	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.13	0.65	2.02	4.05	5.50	5.87	4.98	3.87	2.20	0.93	0.22	0.04	(44)	
	RadStd		0.01	0.07	0.15	0.31	0.38	0.42	0.34	0.22	0.13	0.06	0.01	0.00	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		+0.61	+0.98	+1.80	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (13 neighbors)		+0.46	+0.88	+1.14	+0.46	N/A	N/A	-134	-161	+26	N/A	

Nomenclature: See separate page